Beyond Rest-State, Homogeneous System Identification

Complex initial conditions, non-uniform fields, and a spectral loss, via Differentiable MPM + 3D Gaussian Splatting

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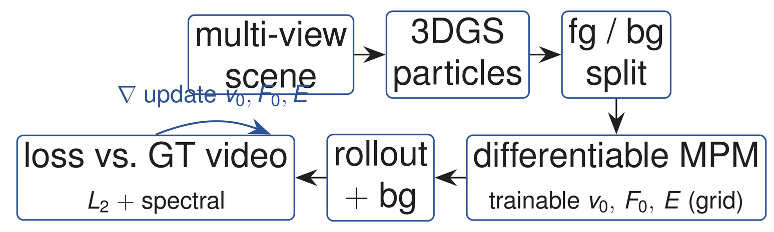
Motivation

Traditional system identification treats an object as a **homogeneous** material at **rest**, moving uniformly in one direction. We pose a harder problem: recover **complex initial conditions on a non-uniform material**. We address four limitations of prior methods:

- **Rest-state assumption.** How can we learn an object *released from a deformed state* at $t=0$?
- **Scalar only.** Can we learn a *field* to model richer setups?
- **Unstable identification.** When is the system actually identifiable?
- **Plain L_2 losses.** They are not good enough for oscillations. We propose a **spectral loss**, and jointly train initial velocity and material while analysing which one dominates.

Method



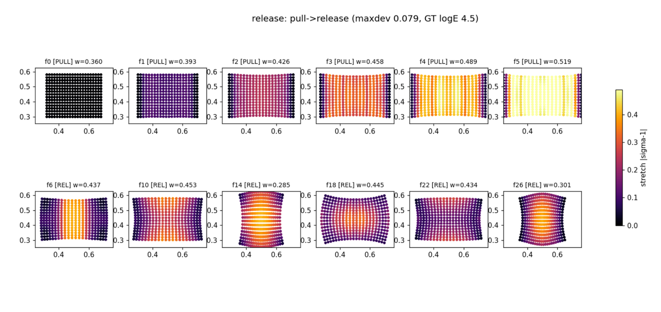
Field trainables use a grid or an MLP queried at the particle position. **Learning F_0.** The rest state X is *unknown*; only the deformed state x_0 is observed. We therefore express the displacement as a function of the *known* x_0, so $X = x_0 - \hat{u}_0(x_0)$ and

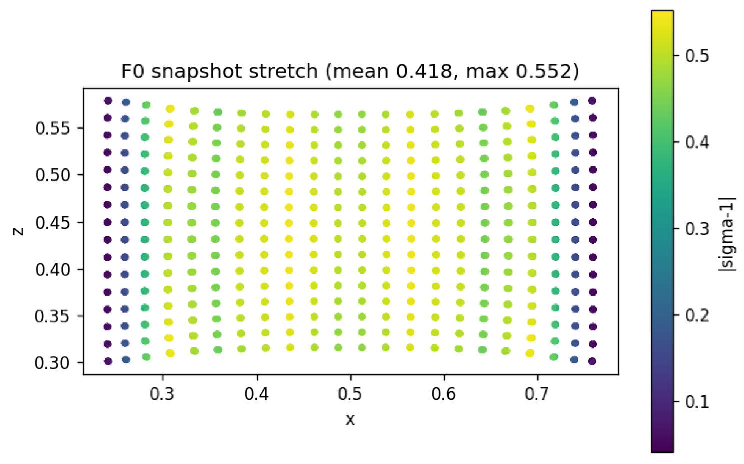
$$F_0 = \frac{\partial x_0}{\partial X} = \left(\frac{\partial X}{\partial x_0}\right)^{-1} = \left(I - \nabla_{x_0}\hat{u}_0\right)^{-1}.$$

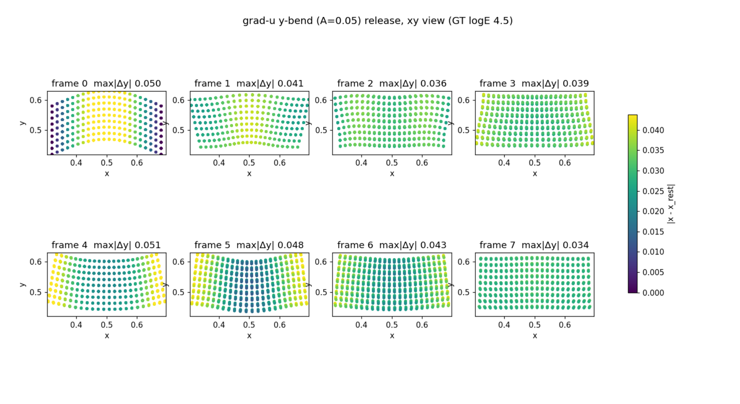
An MLP parametrises $\hat{u}_0(x_0)$; $\nabla_{x_0}\hat{u}_0$ comes from autograd on the known x_0, then we invert.

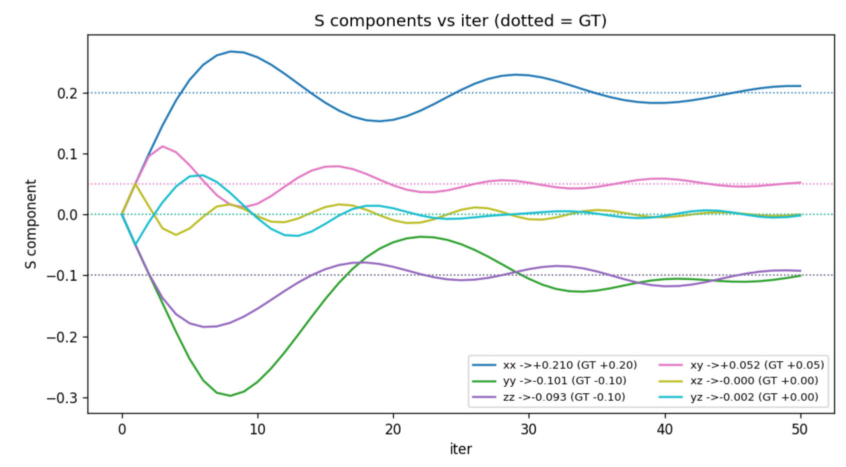
Results: released from a deformed state (F_0)

F_0 is recoverable. A global $V_0 = \exp(S)$ recovers all 6 DOF (max|err| 0.010); a coarse 15-DOF field recovers the target mode.


Per-frame forward (release).

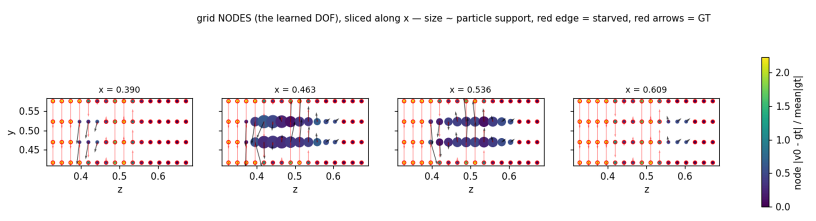

GT vs. predicted stretch.

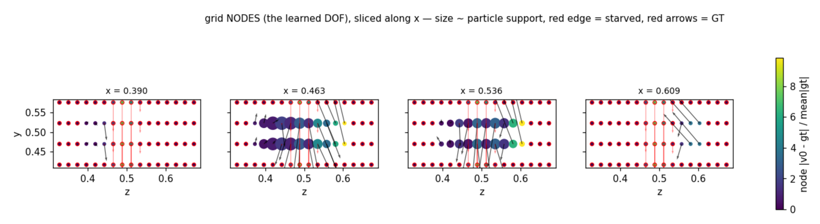

Rest state and $\hat{u}_0$.

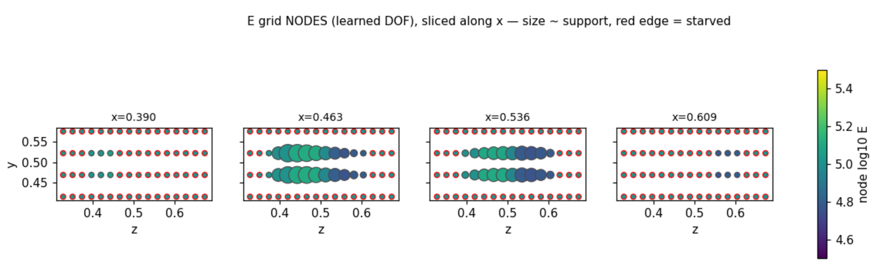

Global S recovery (6 DOF).

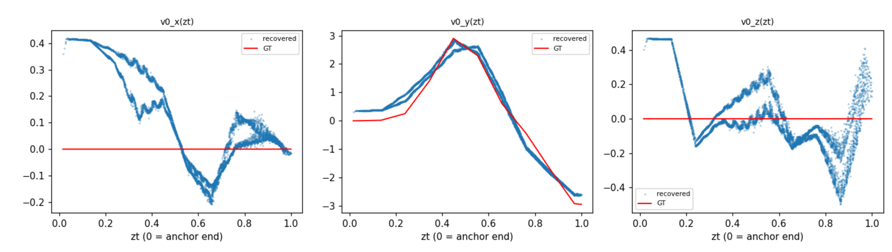
Results: inhomogeneous velocity and material distribution

Non-uniform initial velocity $v_0(\mathbf{x})$ (designed bend / mid-kick excitations) and **non-uniform material** $E(\mathbf{x})$ are recovered as per-particle fields.

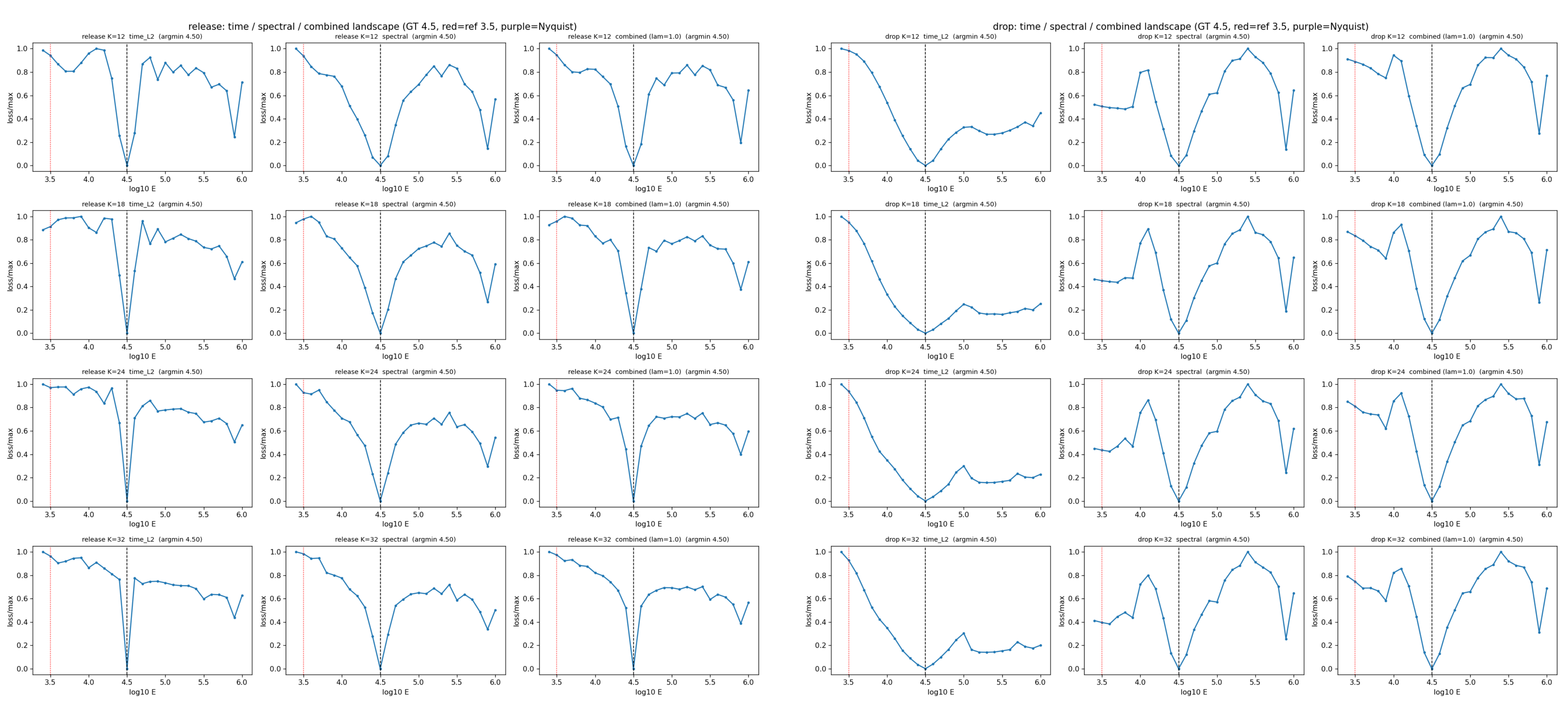

v_0 quiver: bend.


v_0 quiver: mid-kick.


Learned $E(\mathbf{x})$ (ramp-E).


Profile match (GT vs. pred).

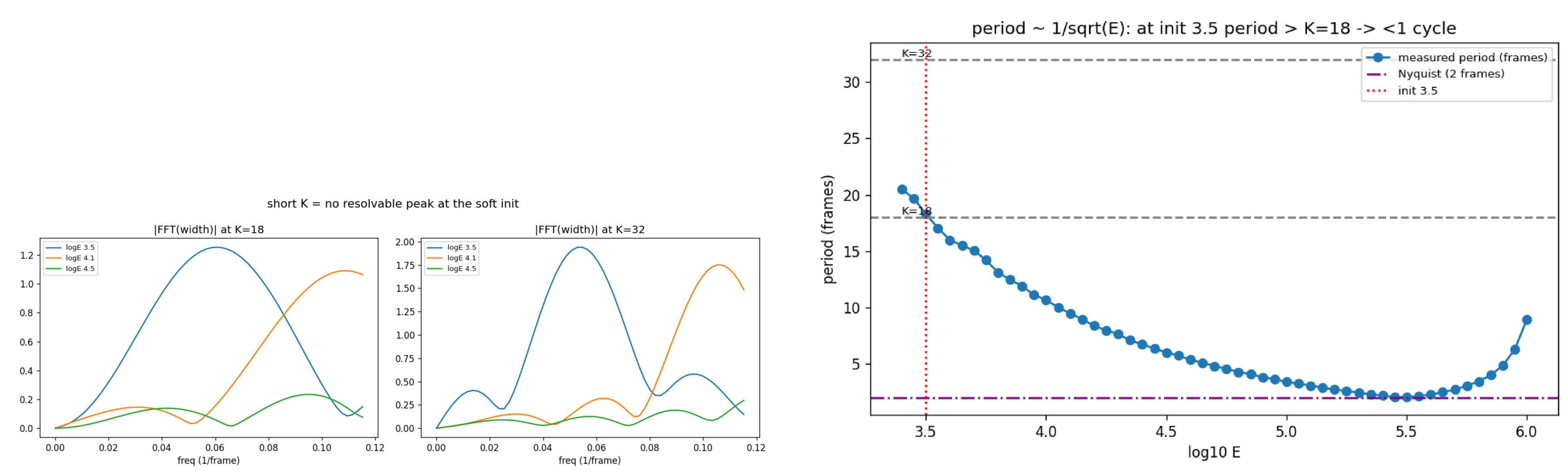
Analysis: amplitude vs. frequency


Two 2D landscapes. Left: E dominant (frequency). Right: v/F dominant on amplitude, E still effective.

Conclusion. E (and ν) controls the **frequency** of motion, while v_0/F_0 only set its **amplitude**. The excitation type therefore decides which channel carries each parameter.

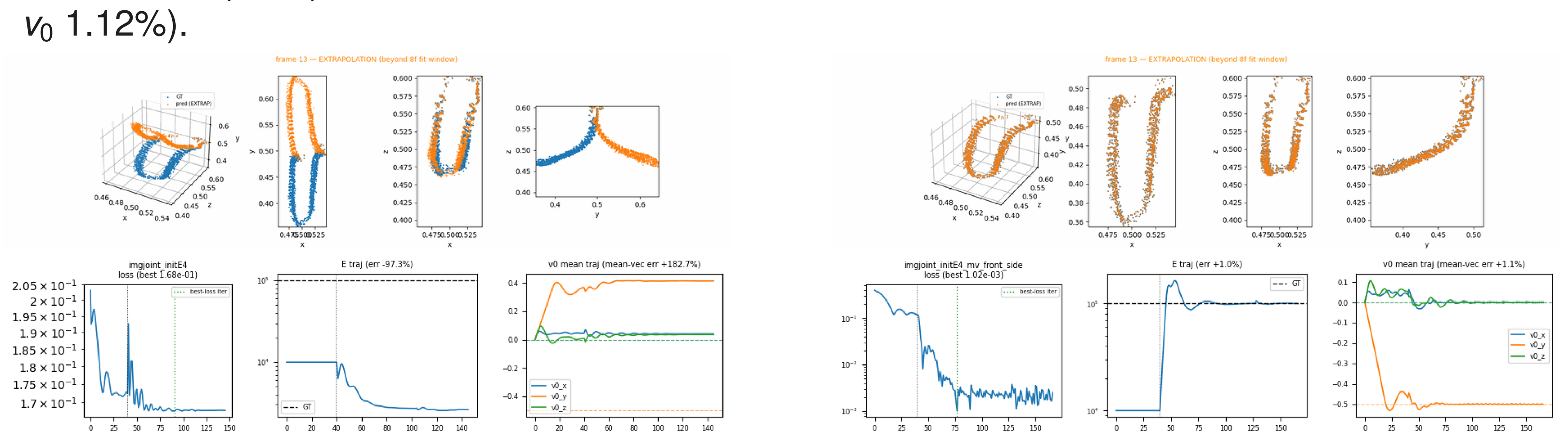
Analysis: a spectral loss for oscillations

When the frequency mismatch is large, a time-domain L_2 loss between two oscillations is **bounded** (distance saturates at $2A$), giving a flat plateau and local minima. A **spectral** loss stays discriminative.


Left: spectra across E. Right: period $\propto 1/\sqrt{E}$, until it hits a Nyquist wall (an aliasing false basin at high E).

Augmenting a non-identifiable scenario

When the excited axis points toward the camera, the initial velocity v_y is hard to learn from a single view: (E, v_0) collapses ($E{\to}2742$, -97%). **Adding a second view resolves it** (E 1.04%, v_0 1.12%).


Single view (left, fails) vs. front + side (right, solved).

Takeaways

1. We recover **complex initial conditions** (released from a deformed state, F_0) and **non-uniform fields** (v_0, E), beyond the rest-state homogeneous assumption.
2. A **spectral loss** reads the frequency channel that a time-domain L_2 loss cannot, with a clear Nyquist limit.
3. E sets frequency, v_0/F_0 set amplitude: excitation design governs identifiability.
4. **Future:** amortise a scene-conditioned physics prior and sample plausible physics (VSD + reconstruction guidance).